02 OCAK Pazartesi
January / Monday
Январь / Понедельник

02.01.2017

02 - Şubat

> 2-363 > 02. Hafta / Week

1st Elimination matrix

1) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{pmatrix} -8r1+r2=r2 \\ mr1+r3=r3 \\ -r1+r4=r4 \end{pmatrix}$

2nd El. matrix

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \rightarrow \begin{matrix} -r2+r3=r3 \\ kr2+r4=r4 \end{matrix}$

3rd El. matrix.

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix} \rightarrow -\frac{r3}{2}+r4=r4$

a) $-8+3\alpha=0 \Rightarrow \alpha=+8/3$ → a 5pts

b) $m+2=0 \Rightarrow m=-2$ → 5pts

$\begin{bmatrix} 1 & +8/3 & -\gamma & +8/3 \\ +8 & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & 64/9 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & +8/3 & -\gamma & -8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & \beta-\frac{16}{3} & 3+2\gamma & \frac{16}{9} \\ 0 & \frac{34}{3} & \gamma-6 & -\frac{2}{3} \end{bmatrix}$

$\frac{49}{3}+\beta-\frac{16}{3}=0 \Rightarrow \boxed{\beta=-11}$

$-\frac{49}{3}\cdot k+\frac{34}{3}=0 \Rightarrow \boxed{k=+\frac{34}{49}}$

Not:

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OCAK Salı
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$\begin{bmatrix} 1 & -8/3 & -\gamma & -8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & -49/3 & 3+2\gamma & -186/9 \\ 0 & 34/3 & \gamma-6 & -2/3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -8/3 & -\gamma & -8/3 \\ 0 & -\frac{49}{3} & -\gamma & -61/3 \\ 0 & 0 & 3+3\gamma & 199/9 \\ 0 & 0 & \gamma-6+\frac{34}{49}\gamma & r \end{bmatrix}$

$r=\frac{199}{9}-\frac{61}{3}\cdot\frac{34}{49}$

$-\frac{3+3\gamma}{2}+\gamma-6-\frac{34}{49}\gamma=0$

$-1.5\gamma-\frac{15}{49}\gamma=7.5 \Rightarrow \gamma=\frac{7.5\cdot 49}{1.5\cdot 59}$

$\gamma=(49/59)\cdot 3$

$\begin{bmatrix} 1 & -8/3 & -49/13 & -8/3 \\ 0 & -49/3 & -49/13 & -61/3 \\ 0 & 0 & 186/13 & 199/9 \\ 0 & 0 & 93/13 & r \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -8/3 & -49/13 & -8/3 \\ 0 & -49/3 & -49/13 & -61/3 \\ 0 & 0 & 186/13 & 199/9 \\ 0 & 0 & 0 & r-\frac{199}{18} \end{bmatrix}$

U

c) the given elimination matrices are enough to find U from A ⇒ there is no other elimination matrices → 10pts

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04.01.2017

04 OCAK Çarşamba
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02. Hafta / Week ➤4-361

2 (a) (10) $x = 5y$ and another geometry

intersection → a volume

$\Rightarrow \begin{pmatrix} x - 5y = 0 \\ \text{another equation} \end{pmatrix}$ must give a volume

$\Rightarrow$ we need 3 free variables $\Rightarrow$ 1 pivot + 3 free

The space must have at least 4 dim.

(an exception) if the equations are same
$\Rightarrow$ whole space can be the intersection.
In this case → the space can be 3-dim

(b) (10) Not
$x - 5y + 0 \cdot z + 0 \cdot k = 0 \iff$
$3x - 2y + 3z + 5k = 9 \iff$ not dependent! to each other

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OCAK Perşembe
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➤02. Hafta / Week ➤5-360

exception if all 3-D is the intersection.

$x - 5y + 0 \cdot z = 0$
$x - 5y + 0 \cdot z = 0$ $\Rightarrow$ the slope intersects itself

(c) (10)
$x - 5y = 0$
$x + 4y - z = 6 \Rightarrow \begin{bmatrix} 1 & -5 & 0 \\ 1 & 4 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$

$\begin{bmatrix} 1 & -5 & 0 \\ 0 & 9 & -1 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 6 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & -5/9 \\ 0 & 9 & -1 \end{bmatrix}$ $\begin{bmatrix} 30/9 \\ 6 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & -5/9 \\ 0 & 1 & -1/9 \end{bmatrix}$ $\begin{bmatrix} 30/9 \\ 6/9 \end{bmatrix}$

P P F

Not: $N(A) = \begin{bmatrix} +5/9 \\ +1/9 \\ 1 \end{bmatrix} \cdot \alpha$, $x_p = \begin{bmatrix} 30/9 \\ 6/9 \\ 0 \end{bmatrix}$

(5) $\Rightarrow x_c = N(A) + x_p = \begin{bmatrix} 10/3 \\ 2/3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5/9 \\ 1/9 \\ 1 \end{bmatrix} \cdot \alpha$

* the solution space is Not a vector space since zero vector is not included
* the solution space is (10) in $\mathbb{R}^3$

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06 OCAK Cuma January / Friday Январь / Пятница

02. Hafta / Week

3) $H^2 = I$ (a)

(5) $H^2 = (I - 2uu^T)\cdot(I - 2uu^T)$

$= (I - 2uu^T - 2uu^T + 4\cdot uu^T uu^T)$

$= I - 4uu^T + 4uu^T = I$ (1 since $u^T u = 1$, (u is unit vector))

(b) if $H^T \cdot H$ is identity $\Rightarrow$ H can be called as orthonormal.

(5) $\Rightarrow H^T \cdot H = (I - 2uu^T)^T \cdot (I - 2uu^T)$

$= (I^T - 2uu^T)(I - 2uu^T)$

$= I - 2uu^T - 2uu^T + 4uu^T uu^T$ ($u^T u = 1$)

$H^T H = I$

(c) $H \cdot u = \lambda \cdot u$ if u is eig. vector.

(5) $H \cdot u = (I - 2uu^T)\cdot u = u - 2uu^T u$ ($u^T u = 1$)

$\Rightarrow H \cdot u = -u \Rightarrow \boxed{\lambda = -1}$

(d) diagonalizable condition $\Rightarrow$

(5) matrix $H_{n\times n}$ must have n indep. eig. vectors

07 OCAK Cumartesi January / Saturday Январь / Суббота

02. Hafta / Week

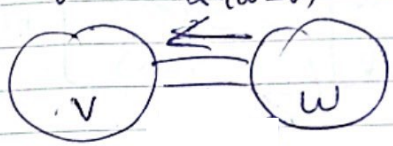
$H_{n\times n} = I_{n\times n} - 2\underbrace{u_{n\times 1}}\cdot\underbrace{u^T_{1\times n}}$

identity. Symmetric matrix

$\Rightarrow H_{n\times n}$ is asymmetric matrix.

$\Rightarrow$ matrix has 'n' eig. values, (different).

$\Rightarrow$ n eig. vectors can be found.

4) 

$\Rightarrow \frac{dw}{dt} = \alpha\cdot(w - v)$

a) $\frac{dv}{dt} = \alpha\cdot(v - w)$

(10) $\Rightarrow \frac{d}{dt}\begin{bmatrix} w \\ v \end{bmatrix} = \alpha\cdot\underbrace{\begin{bmatrix} -1 & +1 \\ +1 & -1 \end{bmatrix}}_{A}\cdot\begin{bmatrix} w \\ v \end{bmatrix}$

b) $|A - \lambda I| = \begin{vmatrix} -1-\lambda & +1 \\ +1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow (-1-\lambda)^2 = 1$ → $\lambda_1 = -2$, $\lambda_2 = 0$

$\lambda_1 = -2 \Rightarrow (A - \lambda I)X_1 = 0 \Rightarrow \begin{bmatrix} +1 & +1 \\ +1 & +1 \end{bmatrix}\cdot X_1 = 0 \Rightarrow X_1 = \begin{bmatrix} +1 \\ -1 \end{bmatrix}$

$\lambda_2 = 0 \Rightarrow X_2 = \begin{bmatrix} 1 \\ +1 \end{bmatrix}$

(5)

08 OCAK Pazar January / Sunday Январь / Воскресенье

02. Hafta / Week

$\begin{bmatrix} w \\ v \end{bmatrix} = c_1\cdot e^{0\cdot t}\cdot\begin{bmatrix} 1 \\ +1 \end{bmatrix} + c_2\cdot e^{-2t}\cdot\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

if $t=0 \Rightarrow \begin{pmatrix} w_0 \\ v_0 \end{pmatrix} = \begin{pmatrix} 40 \\ 10 \end{pmatrix} = c_1\cdot\begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2\cdot\begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow c_1 = 25,\ c_2 = 15$

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09 OCAK
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▸ 9.356 ▸ 03. Hafta / Week

$$\begin{bmatrix} w \\ v \end{bmatrix} = 25 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 15 \cdot e^{-2t} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(5)

if $t = 1\,sec. \Rightarrow \begin{bmatrix} w \\ v \end{bmatrix}_{t=1} = \begin{bmatrix} 25 + 15e^{-2} \\ 25 - 15e^{-2} \end{bmatrix}$

c) $t \to \infty \Rightarrow \begin{bmatrix} w \\ v \end{bmatrix}_{t\to\infty} = \begin{bmatrix} 25 \\ 25 \end{bmatrix} + 15e^{-2t} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ → 0

(10)

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